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FINAL REVIEW

**MULTI-TRAIT HUMAN PROFILING FROM
FINGERPRINTS USING DEEP LEARNING MODELS**

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INTRODUCTION

- ✓ Normal methods to find blood group and gender take time and need physical samples.
- ✓ Fingerprints are easy to collect and unique, so they're a better option.
- ✓ This project predicts blood group, finger type, and gender using fingerprint patterns.
- ✓ It helps in fast identification for forensics, emergency healthcare, and security systems.



Fig 1. Fingerprint

LITERATURE SURVEY

S.NO	AUTHOR	TITLE	JOURNAL DETAILS WITH YEAR	SIGNIFICANT CONTRIBUTION
1	T. R. Prasad. <i>Et al.</i>	Blood Group Detection Using Fingerprint Images	Parul University International Conference on Engineering and Technology 2025 (PiCET 2025).	• Inference: CNN achieved 92% test accuracy for ABO/Rh. • Problem Identified: Fingerprint–blood-group link is weak; larger datasets needed.
2	C. Sivamurugan <i>et al.</i>	Enhanced Blood Group Prediction with Fingerprint Images using Deep Learning,	<i>Proc. 3rd Int. Conf. Automation, Computing and Renewable Systems (ICACRS 2024)</i> , 2024, pp. 1091–1094	• Inference: VGG-16-based transfer learnin with train accuracy of 98% and test accuracy of 87.86%. • Problem Identified: suffers over-fitting on limited finger-blood datasets
3	N. Kapoor, A. Badiye and S. D. Mishra <i>et al.</i>	Fingerprint Analysis for the Determination of Hand Origin Using the Axis Slant in Whorl Patterns	<i>Forensic Sciences Research</i> , vol. 7, no. 2, pp. 1-5, 2020	• Inference: Whorl-axis slant identifies left/right hand with $\approx$ 81 % directional accuracy • Problem Identified: method applies only to whorl prints; loops/arches need alternate cues

LITERATURE SURVEY

S.NO	AUTHOR	TITLE	JOURNAL DETAILS WITH YEAR	SIGNIFICANT CONTRIBUTION
4	Y. A. Al-Wajih <i>et al.</i>	Finger Type Classification with Deep Convolution Neural Networks	<i>Int. Conf. Informatics in Control, Automation and Robotics (ICINCO 2022)</i> , Lisbon, Portugal, 2022, pp. 247–254.	• Inference: CNN classifies ten finger positions at ~94 % accuracy using public datasets • Problem Identified: cross-sensor domain shift still degrades performance
5	J. Serin <i>et al.</i>	Gender Classification from Fingerprint Using Hybrid CNN-SVM	<i>J. Artificial Intelligence & Technology</i> , vol. 4, no. 1, pp. 82-87, 2024	• Inference: Hybrid CNN-SVM reached 99.25 % accuracy for gender prediction • Problem Identified: Research still lacks large, diverse datasets for real-world deployment.
6.	Lubana Akter <i>et al.</i>	Comparative Analysis of Fingerprint-Based Gender Classification Employing Convolutional Neural Network	PICET Conference (IEEE), pp. 1–5, 2024	• Inference: CNN achieved 99.86% accuracy for fingerprint-based gender classification. • Problem Identified: Performance may drop with poor-quality images or real-world noise; dataset diversity is still limited.



OBJECTIVE



- To develop a machine learning system that predicts blood group, gender and finger type by analyzing the fingerprint patterns.
- To establish a non-invasive, pattern-based approach for multi-attribute biometric classification and screening.

LEGACY APPROACHES TO MULTI-TRAIT DETECTION USING FINGERPRINT ANALYSIS

BLOOD GROUP :

- CNN models (AlexNet, LeNet, VGG16, ResNet34) tested on SOCOFing + NIST SD4 with an 80/20 split and Adam showed strong performance, with **ResNet34** hitting **92%** accuracy for fast, non-invasive blood-group identification.

GENDER :

- A CNN model trained on the SOCOFing dataset (55,273 images) using conv layers, GAP, dropout, and FC layers achieved **99.86%** gender-classification accuracy, outperforming KNN and SVM.

FINGER TYPE:

- SVM models using ridge-density and minutiae-dispersion features were tested on 3,500 fingerprint images, achieving **88.9%** accuracy for reliable finger-type prediction.

BLOCK DIAGRAM

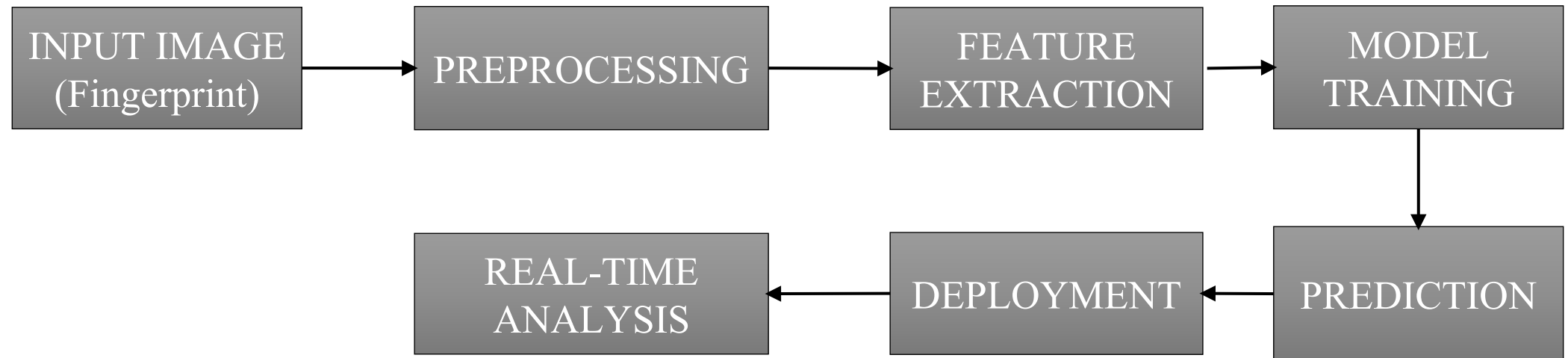


Fig 2. Prediction Pipeline

DATASET SPECIFICATIONS

Table 1. Datasets

DATASET NAME	SIZE	CLASSES
Fingerprints Blood group	8000	8
SOCOFing – Gender & Finger-Type	55,000	4

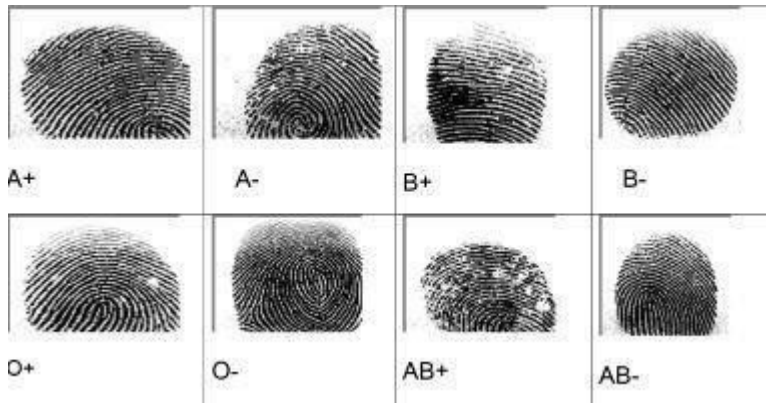


Fig 3. Blood group dataset

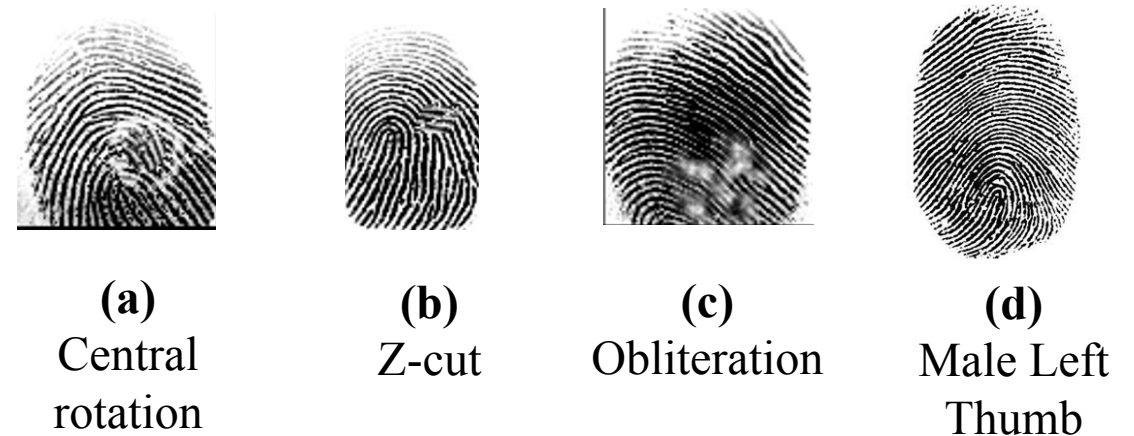


Fig 4. Finger-type & Gender dataset

FEATURE EXTRACTION

Table 2. Feature Extraction

LEVEL	FEATURES
Global level features	Loops, Whorls, Arches
Minutiae level features	Ridges, Core, Dot, Bifurcation

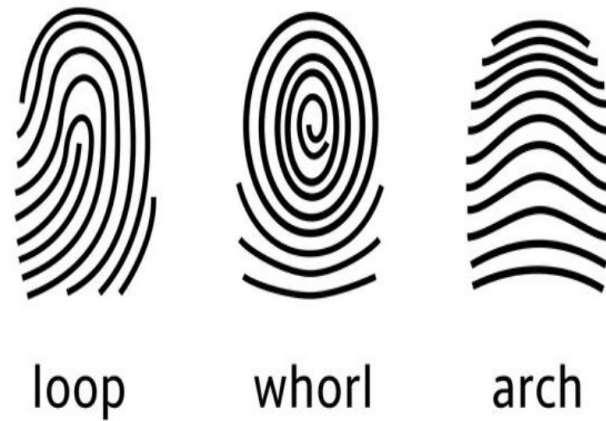


Fig 5. Global Level Features

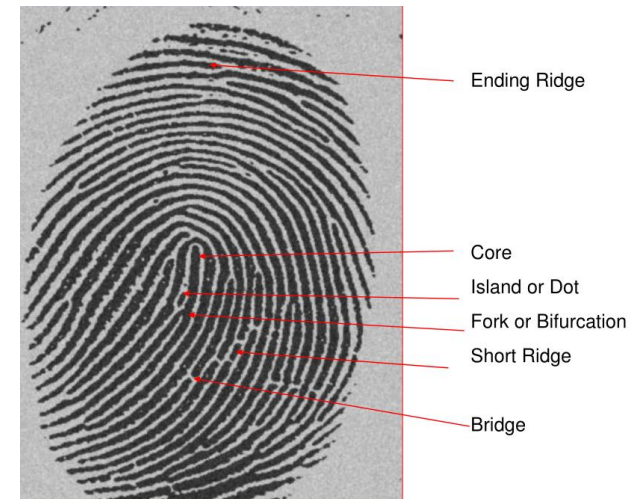


Fig 6. Minutiae Level Features

PROPOSED MODEL ARCHITECTURE

CNN [22 Layers]

Layer (type)	Output Shape	Param #
conv2d_6 (Conv2D)	(None, 128, 128, 32)	896
batch_normalization_7 (BatchNormalization)	(None, 128, 128, 32)	128
conv2d_7 (Conv2D)	(None, 128, 128, 32)	9,248
batch_normalization_8 (BatchNormalization)	(None, 128, 128, 32)	128
max_pooling2d_3 (MaxPooling2D)	(None, 64, 64, 32)	0
dropout_4 (Dropout)	(None, 64, 64, 32)	0
conv2d_8 (Conv2D)	(None, 64, 64, 64)	18,496
batch_normalization_9 (BatchNormalization)	(None, 64, 64, 64)	256
conv2d_9 (Conv2D)	(None, 64, 64, 64)	36,928
batch_normalization_10 (BatchNormalization)	(None, 64, 64, 64)	256
max_pooling2d_4 (MaxPooling2D)	(None, 32, 32, 64)	0
dropout_5 (Dropout)	(None, 32, 32, 64)	0
conv2d_10 (Conv2D)	(None, 32, 32, 128)	73,856
batch_normalization_11 (BatchNormalization)	(None, 32, 32, 128)	512
conv2d_11 (Conv2D)	(None, 32, 32, 128)	147,584
batch_normalization_12 (BatchNormalization)	(None, 32, 32, 128)	512
max_pooling2d_5 (MaxPooling2D)	(None, 16, 16, 128)	0
dropout_6 (Dropout)	(None, 16, 16, 128)	0
flatten_1 (Flatten)	(None, 32768)	0
dense_2 (Dense)	(None, 256)	8,388,864
batch_normalization_13 (BatchNormalization)	(None, 256)	1,024
dropout_7 (Dropout)	(None, 256)	0
dense_3 (Dense)	(None, 8)	2,056

Fig 7. Blood group Model

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 224, 224, 32)	832
conv2d_1 (Conv2D)	(None, 224, 224, 32)	25632
max_pooling2d (MaxPooling2D)	(None, 112, 112, 32)	0
dropout (Dropout)	(None, 112, 112, 32)	0
conv2d_2 (Conv2D)	(None, 112, 112, 64)	18496
conv2d_3 (Conv2D)	(None, 112, 112, 64)	36928
max_pooling2d_1 (MaxPooling2D)	(None, 56, 56, 64)	0
dropout_1 (Dropout)	(None, 56, 56, 64)	0
conv2d_4 (Conv2D)	(None, 56, 56, 128)	73856
conv2d_5 (Conv2D)	(None, 56, 56, 128)	147584
max_pooling2d_2 (MaxPooling2D)	(None, 28, 28, 128)	0
dropout_2 (Dropout)	(None, 28, 28, 128)	0
flatten (Flatten)	(None, 100352)	0
dense (Dense)	(None, 100)	10035300
dropout_3 (Dropout)	(None, 100)	0
dense_1 (Dense)	(None, 20)	2020

Total params: 10,340,648
Trainable params: 10,340,648
Non-trainable params: 0

Fig 8. Gender & Finger-type Model

TRAINING SETUP COMPARISON

Table 3. Training Setup

Parameters	Blood group Model	Gender & Finger-Type Model
Split Ratio (Train : Val : Test)	80 : 10 : 10	80 : 10 : 10
Total Images (Train : Val : Test)	6400 : 800 : 800	15,152 : 3,788 : 2,460
Augmentation	No	Undersampled
Optimizer	Adam (0.001)	Adam (0.0001)
Activation function	ReLu + Softmax	ReLu + Softmax
Loss function	Categorical Crossentropy	Categorical Crossentropy

PERFORMANCE COMPARISON

Table 4. Blood group detection

Model Name	Train %	Val %	Test %
CNN (22 Layer)	97.8	91	91.1
CNN (GeLu)	87.48	89.08	90.62
VGG-16	93.85	90.56	88.16
ResNet	90	88	87
InceptionV3	100	99.98	70
EfficientNet B0	99.21	86.33	80.75
EfficientNet B1	90.68	92.6	80
EfficientNet B3	91.72	90.00	82

Table 5.Gender & Finger-Type Prediction

Model Name	Train%	Val %	Test %
CNN (22 Layer)	96.71	95.92	93.80
CNN (28 Layer)	100	97.16	96.86
CNN (16 Layer)	91.04	90.12	94.51
CNN (8 Layer)	81.70	77.55	75.43
MobileNet	72	49.89	48.12
DenseNet121	94.75	90.04	89.42
InceptionV3	99	72	72
EfficientNet B4	96.80	88.25	87.90

Inference : The custom **22-layer CNN** delivers the highest accuracy for both tasks, while pre-trained models like EfficientNet, MobileNet, and InceptionV3 struggle with fingerprint-specific patterns. Overall, the **lightweight** custom CNN generalizes better and provides more stable, reliable results.

PERFORMANCE COMPARISON BETWEEN EXISTING STUDIES AND PROPOSED APPROACH

Table 6. Blood group detection

AUTHOR	MODEL USED	ACCURACY%
Proposed model (2025)	CNN	97.8
Bhat et al. (2025)	VGG16	90
Ram et al. (2025)	CNN	94.2
Prasad et al. (2025)	ViT	92.5
Rao et al. (2024)	CNN	89.4
Sharma et al. (2024)	KNN	88

Table 7.Gender & Finger-Type Prediction

AUTHOR	MODEL USED	ACCURACY%
Proposed model (2025)	CNN	96.71
Dutta et al. (2025)	SVM (Finger)	88.9
Aktar <i>et al.</i> (2024)	CNN (Gender)	99.86
Mensa et al. (2024)	CNN (Gender)	90.4
Hussain et al. (2023)	VGG16 (Gender)	85.7
Oladele et al. (2022)	CNN (Gender)	72

Inference : For blood-group detection, earlier models like VGG16, ViT, KNN, and standard CNNs achieved up to **94%**, while the proposed approach reaches **97.8%**, showing better feature learning from fingerprint patterns. For gender and finger-type prediction, previous works ranged from **72% to 99.86%** (gender-only), but the proposed multi-trait CNN achieves **96.71%**, proving it is more robust and reliable across combined tasks compared to single-trait models.

PERFORMANCE METRICS

1. Blood Group Prediction [CNN 22 Layers]

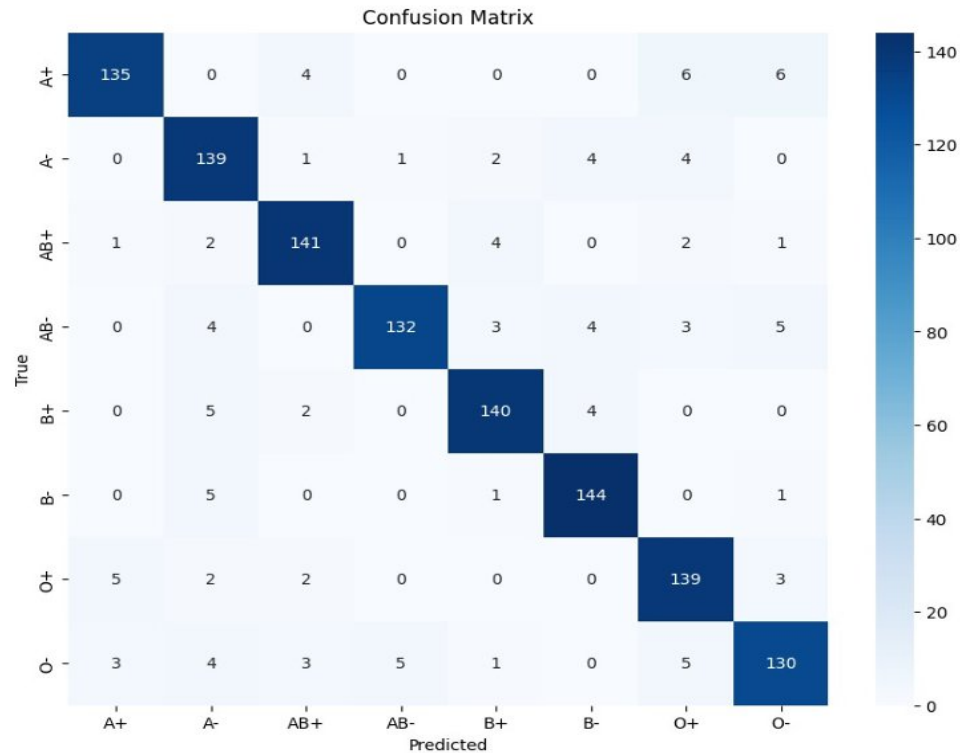


Fig 9. Confusion Matrix

	precision	recall	f1-score	support
A+	0.94	0.89	0.92	151
A-	0.86	0.92	0.89	151
AB+	0.92	0.93	0.93	151
AB-	0.96	0.87	0.91	151
B+	0.93	0.93	0.93	151
B-	0.92	0.95	0.94	151
O+	0.87	0.92	0.90	151
O-	0.89	0.86	0.88	151
accuracy			0.91	1208
macro avg	0.91	0.91	0.91	1208
weighted avg	0.91	0.91	0.91	1208

Fig 10. Metrics Evaluation

2. Gender & Finger-Type Prediction [CNN 22 Layers]

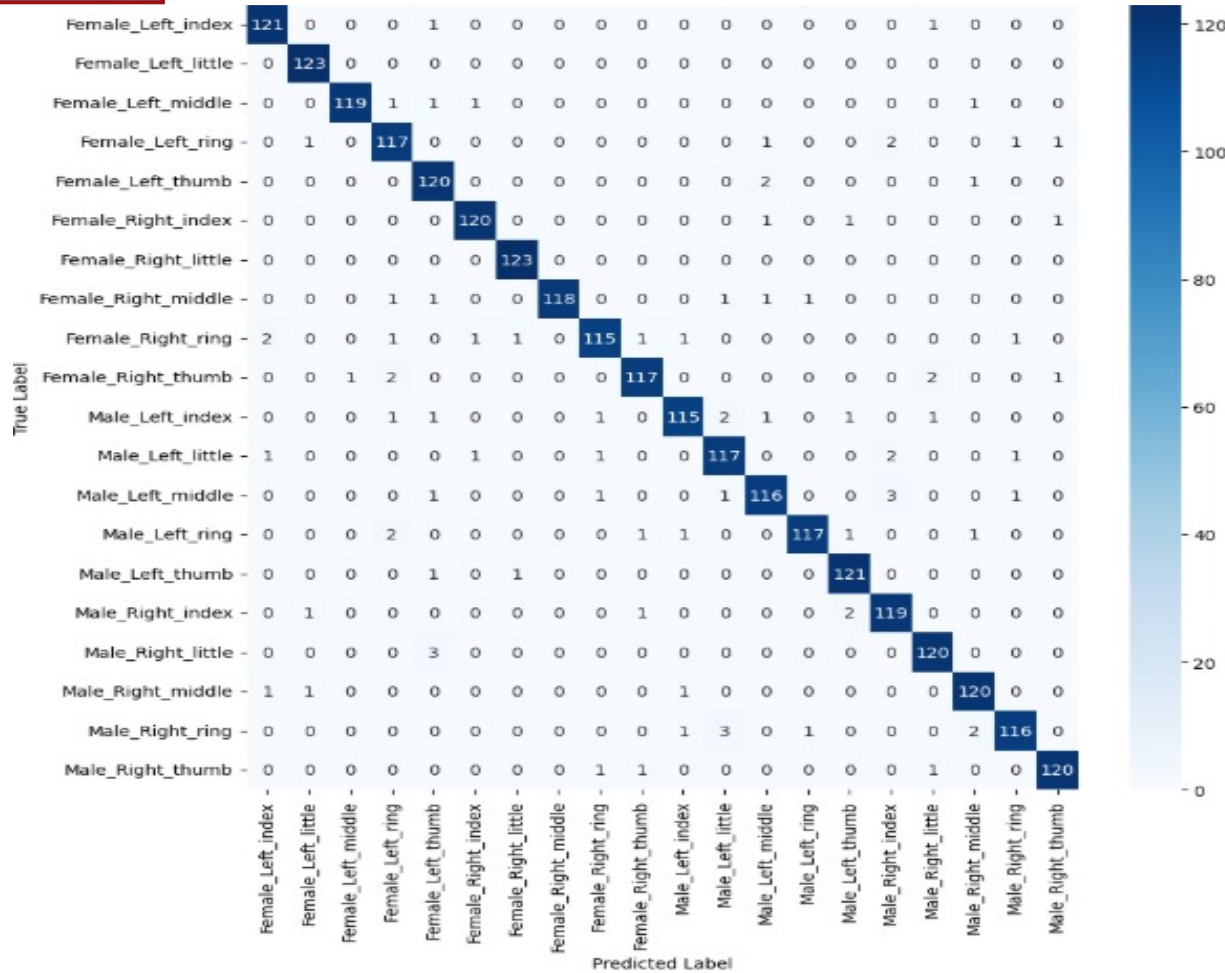


Fig 11. Confusion Matrix

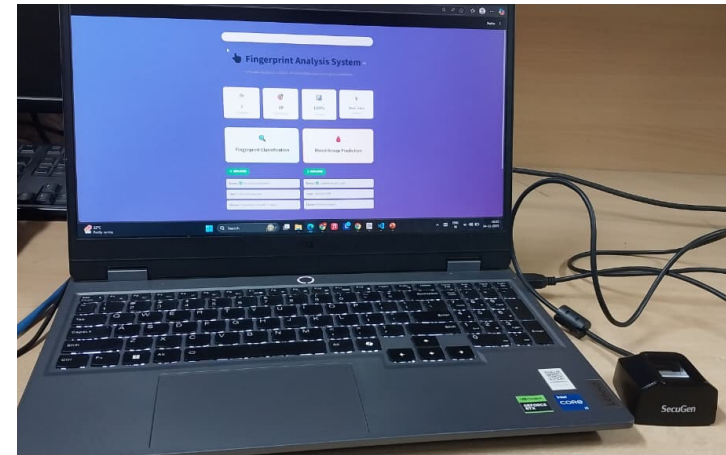
	precision	recall	f1-score	support
Female Left index	0.98	1.00	0.99	123
Female Left little	0.99	1.00	1.00	123
Female Left middle	1.00	1.00	1.00	123
Female Left ring	1.00	1.00	1.00	123
Female Left thumb	0.98	1.00	0.99	123
Female Right index	0.98	1.00	0.99	123
Female Right little	0.98	1.00	0.99	123
Female Right middle	0.98	1.00	0.99	123
Female Right ring	0.98	1.00	0.99	123
Female Right thumb	0.95	1.00	0.97	123
Male Left index	0.92	0.89	0.90	123
Male Left little	0.90	0.89	0.89	123
Male Left middle	0.83	0.86	0.85	123
Male Left ring	0.88	0.85	0.86	123
Male Left thumb	0.96	0.97	0.96	123
Male Right index	0.95	0.85	0.90	123
Male Right little	0.91	0.84	0.87	123
Male Right middle	0.90	0.90	0.90	123
Male Right ring	0.84	0.93	0.88	123
Male Right thumb	1.00	0.93	0.96	123
accuracy			0.95	2460
macro avg	0.95	0.95	0.94	2460
weighted avg	0.95	0.95	0.94	2460

Fig 12. Metrics Evaluation

TECHNIQUE FOR FINGERPRINT COLLECTION



(a)
Ink-Stamp



(b)
SecuGen
Fingerprint Sensor.

Fig 13. Techniques Used

COLLECTED FINGERPRINTS



(a) (b) (c) (d)

Fig 14. Ink-Stamp prints
a) Male Right Thumb (B-),
b) Male Left Index (O+),
c) Female Right Middle (B+),
d) Female Left Thumb (O+).



(a) (b) (c) (d)

Fig 15. Scanned prints
a) Male Right Index (O+),
b) Male Right Ring (O+),
c) Female Right Thumb (B-),
d) Male Left Middle (B-).

TESTING OF THE TRAINED MODELS

```
predict_image('/kaggle/input/finger-print-based-blood-group-dataset/  
dataset_blood_group/0+/cluster_6_102.BMP')
```

```
1/1 _____ 0s 29ms/step  
Predicted: 0 pos
```

```
predict_image('/kaggle/input/finger-print-based-blood-group-dataset/  
dataset_blood_group/A+/cluster_0_1005.BMP')
```

```
Predicted: A_pos  
1/1 _____ 0s 30ms/step
```

```
test_image_path2 = '/kaggle/input/socofing/SOCOFing/Altered/Altered-Medium/103__F_Left_ring_finge  
r_0b1.BMP'
```

```
Predicted Finger Type: Female_Left_ring
```

```
test_image_path3 = '/kaggle/input/socofing/SOCOFing/Real/108__M_Right_thumb_finger.BMP'
```

```
Predicted Finger Type: Male_Right_thumb
```

Fig 16. Model testing (Datasets from online)

DEPLOYMENT

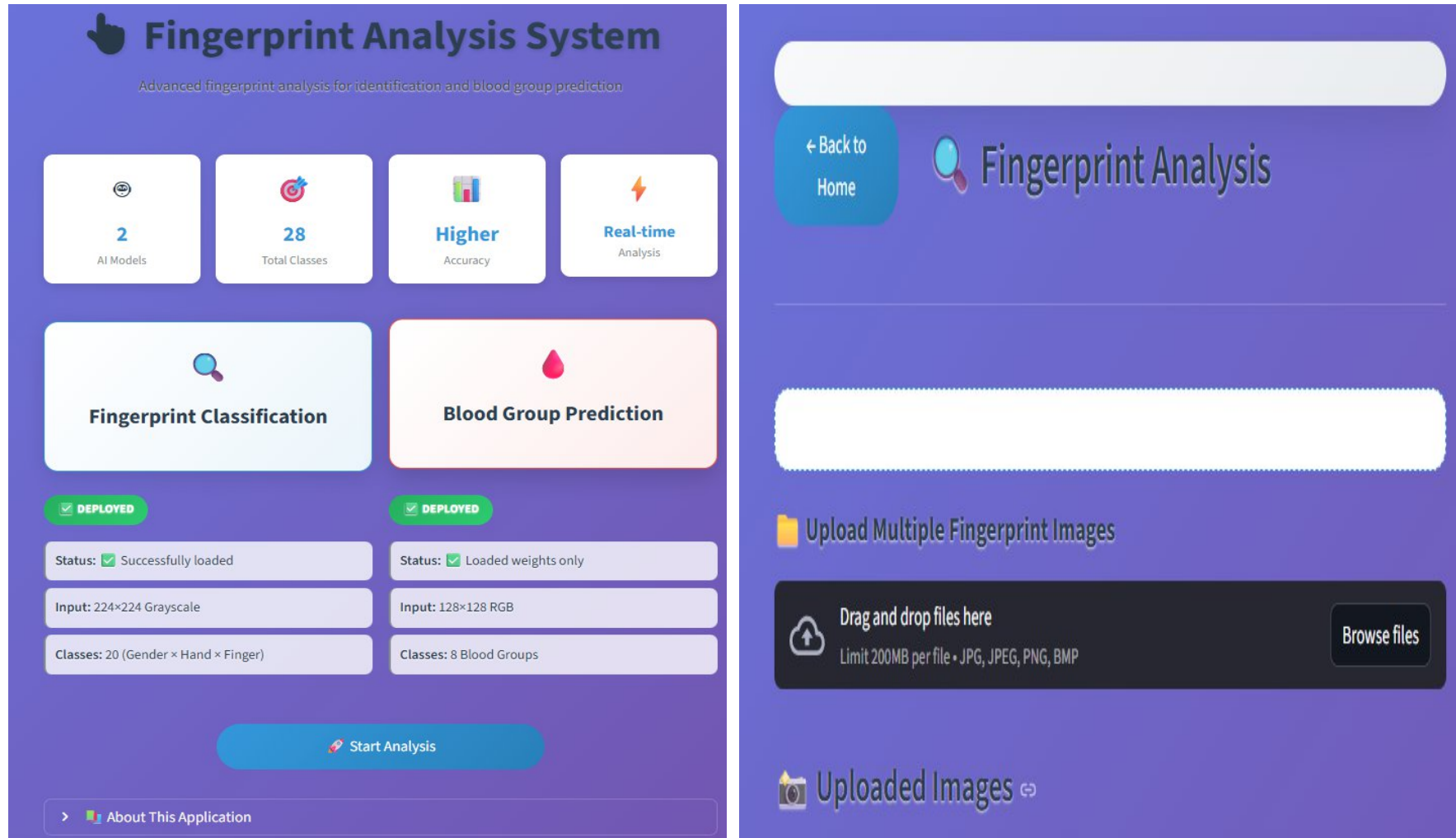


Fig 17. Deployment Interface (Streamlit)

TESTING

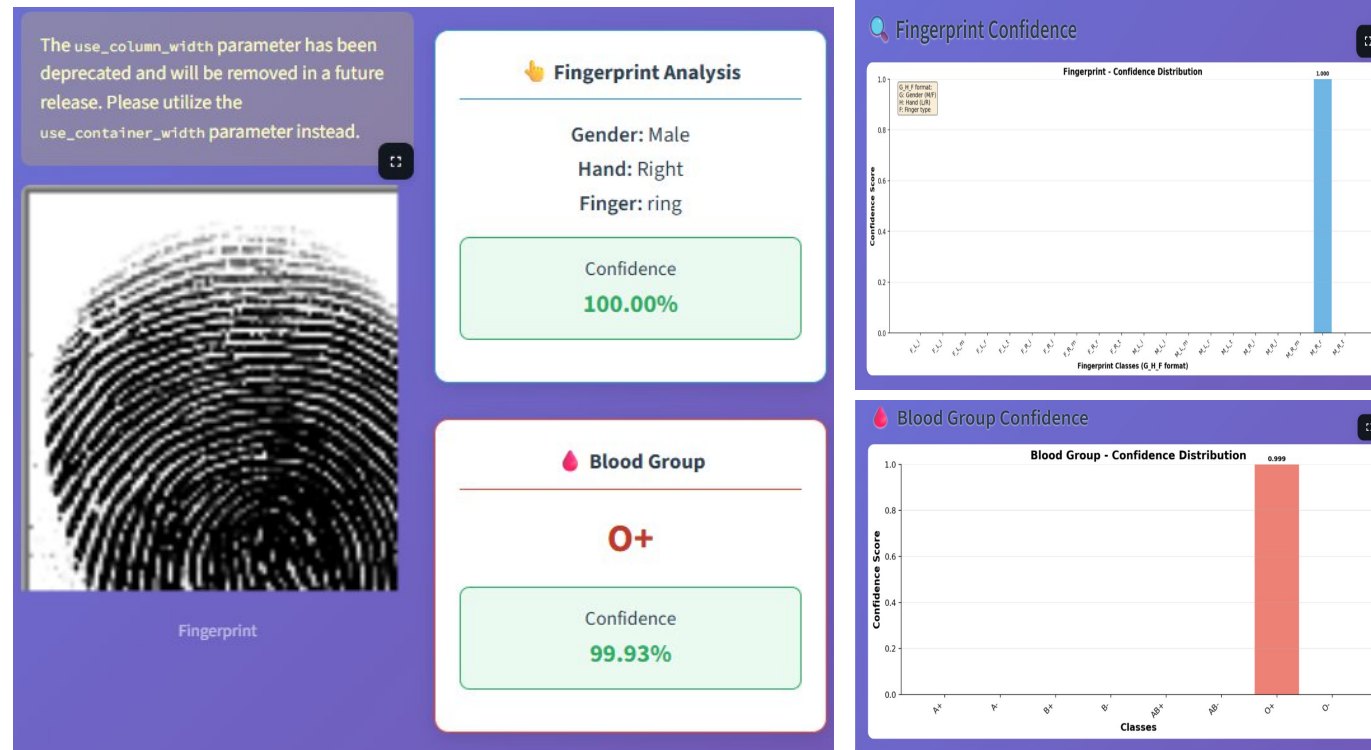


Fig 18. Prediction of Male Right Ring Finger with O+ blood group (Online datasets)

Key observation: The 100% confidence is due to clean online datasets; real-world noisy fingerprints may reduce this accuracy.

PREDICTION FROM SCANNED FINGERPRINTS

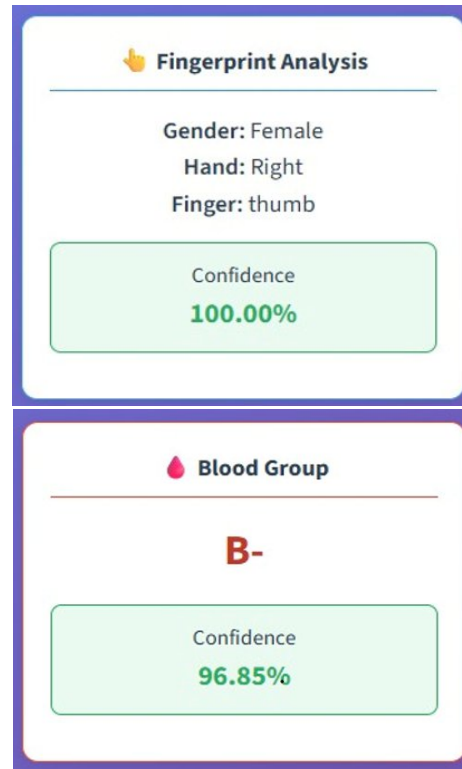


Fig 19. Female Right Thumb
Bloodgroup : B-

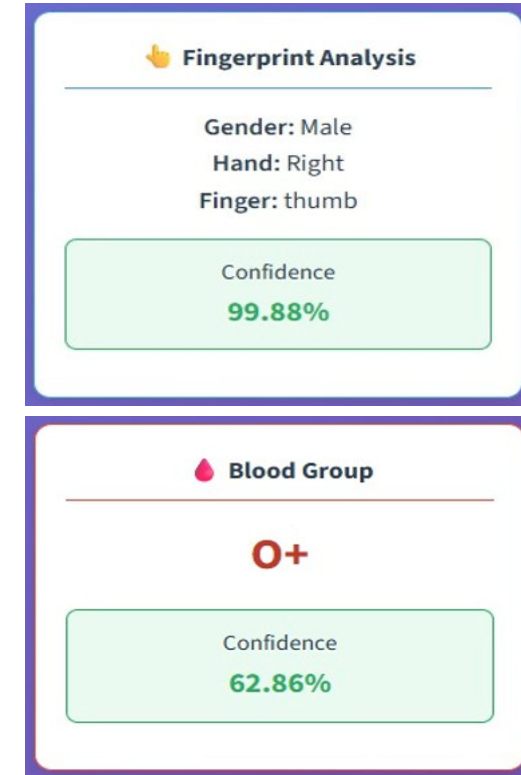


Fig 20. Male Right Thumb
Bloodgroup : O+

PREDICTION FROM SCANNED FINGERPRINTS



 **Fingerprint Analysis**

Gender: Male
Hand: Right
Finger: ring

Confidence
99.98%

 **Blood Group**

O-

Confidence
69.12%

Fig 21. Male Right Ring
Bloodgroup : O+



 **Fingerprint Analysis**

Gender: Male
Hand: Right
Finger: index

Confidence
86.33%

 **Blood Group**

B-

Confidence
69.52%

Fig 22. Male Right Index
Bloodgroup : B-

PREDICTION FROM SCANNED FINGERPRINTS



👉 Fingerprint Analysis

Gender: Male
Hand: Right
Finger: middle

Confidence
83.02%

📌 Blood Group

O+

Confidence
84.88%

Fig 23. Male Right Middle
Bloodgroup : O+



👉 Fingerprint Analysis

Gender: Male
Hand: Left
Finger: middle

Confidence
99.58%

📌 Blood Group

B-

Confidence
99.27%

Fig 24. Male Left Middle
Bloodgroup : B-

PREDICTION FROM STAMP FINGERPRINTS



Fingerprint Analysis

Gender: Male
Hand: Right
Finger: thumb

Confidence
49.66%

Blood Group

B-

Confidence
99.28%

Fig 25. Male Right Thumb
Bloodgroup : B-



Fingerprint Analysis

Gender: Male
Hand: Right
Finger: ring

Confidence
99.74%

Blood Group

AB-

Confidence
43.59%

Fig 26. Male Right Ring
Bloodgroup : O+

PREDICTION FROM STAMP FINGERPRINTS



Fingerprint Analysis

Gender: Male
Hand: Right
Finger: ring

Confidence
87.87%

Blood Group

B-

Confidence
99.87%

Fig 27. Male Right Ring
Bloodgroup : O+



Fingerprint Analysis

Gender: Male
Hand: Right
Finger: middle

Confidence
83.02%

Blood Group

O-

Confidence
69.12%

Fig 28. Male Right Index
Bloodgroup : O+



```

499 def create_simplified_fingerprint_chart(predictions, class_names)
503     for name in class_names:
504         parts = name.split('_')
505         if len(parts) >= 3:
506             simplified = f"{parts[0][0]}_{parts[1][0]}_{parts[2]
507             simplified_names.append(simplified)
508         else:
509             simplified_names.append(name)
510
511     fig, ax = plt.subplots(figsize=(14, 6))
512
513     # Create bars
514     bars = ax.bar(range(len(simplified_names)), predictions, col
515
516     # Customize the chart
517     ax.set_xlabel('Fingerprint Classes (G_H_F format)', fontsize
518     ax.set_ylabel('Confidence Score', fontsize=11, fontweight='b
519     ax.set_title(f'{title} - Confidence Distribution', fontsize=
520
521     # Set x-axis ticks to simplified names
522     ax.set_xticks(range(len(simplified_names)))
523     ax.set_xticklabels(simplified_names, rotation=45, ha='right'
524
525     # Add value labels on top of bars

```

CONCLUSION

- This project proves that deep learning can accurately predict blood group, gender, and finger-type from fingerprint images.
- Using a custom 22-layer CNN and real-time fingerprint sensor integration, the system achieved **97.8%** accuracy for blood groups and **96.71%** for gender and finger type.
- The live sensor connection allows instant capture and processing, removing the need for manual uploads.
- Overall, it makes the system fast, non-invasive, and suitable for real-time field applications.



FUTURE WORK

- **Advanced Sensor Technology:** Deploy multi-spectral, high-resolution fingerprint scanners in clinics, emergency rooms, and crime scenes to capture clean ridge patterns even from dry, wet, or damaged fingers.
- **Multi-Modal Biometric Fusion:** Integrate fingerprint analysis with iris, face, or palm-vein recognition for use in border security, high-security access control, forensic identification, and disaster victim profiling.
- **Portable & Secure Edge Deployment:** Use handheld wireless fingerprint devices embedded with edge AI and blockchain-based secure logging for on-site verification in ambulances, remote medical camps, police field units, and mobile identity checkpoints.

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THANK YOU